deepSSF Data Prep

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Here we prepare the data for fitting a deepSSF model. We load the GPS data and tidy it, create a trajectory object containing a series of steps and read in the environmental layers. For each observed step we crop out local subsets of the environmental layers, centred on the step's location. For every step we will then have the surrounding environmental information stored as a stack of small rasters, with the 'target' of what we're trying to predict (the actual location of the next step) as a raster layer as well, with all cells being 0 except the observed next location, which is 1. These, along with temporal covariates and the previous bearing, will be the input data for the deepSSF model.

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Loading packages

```r
library(tidyverse)
packages <- c("amt", "sf", "terra", "beepr", "tictoc")
walk(packages, require, character.only = T)
```

Import data

```r
buffalo <- read_csv("data/buffalo.csv")
```

```
New names:
Rows: 133161 Columns: 11
-- Column specification
-------------------------------------------------------- Delimiter: "," chr
(3): node, dates, DateTime dbl (7): ...1, lat, lon, height, accuracy, heading,
speed dttm (1): timestamp
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```r
head(buffalo)
```

```
# A tibble: 6 x 11
   ...1 timestamp             lat   lon height accuracy heading speed node
  <dbl> <dttm>              <dbl> <dbl>  <dbl>    <dbl>   <dbl> <dbl> <chr>
1   169 2018-07-25 00:04:02 -12.3  134.   91.3      977     0   0     2005
2   170 2018-07-25 01:04:23 -12.3  134.   92.1      704    77.0 0.034 2005
3   171 2018-07-25 02:04:39 -12.3  134.   95.2     1105   129.  0.008 2005
4   172 2018-07-25 03:04:17 -12.3  134.  104.       865     0   0.013 2005
5   173 2018-07-25 04:04:39 -12.3  134.   85.5      711   231.  0.01  2005
6   174 2018-07-25 05:04:27 -12.3  134.   95.8      780   207.  0.012 2005
# i 2 more variables: dates <chr>, DateTime <chr>
```

Tidy data

```
# remove individuals that have poor data quality or less than about 3 months of data.
# The "2014.GPS_COMPACT copy.csv" string is a duplicate of ID 2024, so we also exclude it
buffalo <- buffalo %>% filter(!node %in% c("2014.GPS_COMPACT copy.csv",
                                           # 2005, 2014, 2018, 2021, 2022, 2024,
                                           2029, 2043, 2265, 2284, 2346, 2354))

# arrange by time and exclude any duplicate timestamps (within the same individual's data)
buffalo <- buffalo %>%
  group_by(node) %>%
  arrange(DateTime, .by_group = T) %>%
  distinct(DateTime, .keep_all = T) %>%
  arrange(node) %>%
  mutate(ID = node)

# keep only the relevant columns
buffalo_clean <- buffalo[, c(12, 2, 4, 3)]

# rename the columns
colnames(buffalo_clean) <- c("id", "time", "lon", "lat")

# convert the time to the correct timezone
attr(buffalo_clean$time, "tzone") <- "Australia/Queensland"
head(buffalo_clean)
```

```
# A tibble: 6 x 4
  id    time                  lon   lat
  <chr> <dttm>              <dbl> <dbl>
1 2005  2019-01-01 11:01:14  134. -12.3
2 2005  2019-01-01 21:00:59  134. -12.3
3 2005  2019-01-01 22:01:15  134. -12.3
4 2005  2019-01-01 23:01:28  134. -12.3
5 2005  2019-01-02 00:01:12  134. -12.3
6 2005  2019-01-02 03:00:36  134. -12.3
```

```
# check timezone
tz(buffalo_clean$time)
```

```
[1] "Australia/Queensland"
```

```r
# create a vector of the unique ids for indexing later
buffalo_ids <- unique(buffalo_clean$id)
```

Create a subset of the data to export for testing the deepSSF package

We'll select the ID 2005, as that was used for training the model in the MEE paper

```r
buffalo_clean_2005 <- buffalo_clean %>% filter(id == 2005)

# convert to sf object and transform to the same projection as the covariates (GDA94 / Geosc
buffalo_clean_2005_sf <- buffalo_clean_2005 %>%
  st_as_sf(coords = c("lon", "lat"), crs = 4326) %>%
  st_transform(crs = 3112)

# add the coordinates back to the data frame
buffalo_clean_2005_coords <- buffalo_clean_2005_sf %>%
  st_coordinates() %>%
  as.data.frame() %>%
  rename(x = X, y = Y)

buffalo_clean_2005_proj <- buffalo_clean_2005 %>%
  bind_cols(buffalo_clean_2005_coords) %>%
  dplyr::select(id, time, x, y)

buffalo_clean_2005_proj %>% write_csv("data/buffalo_djelk_id2005.csv")
```

Setup trajectory

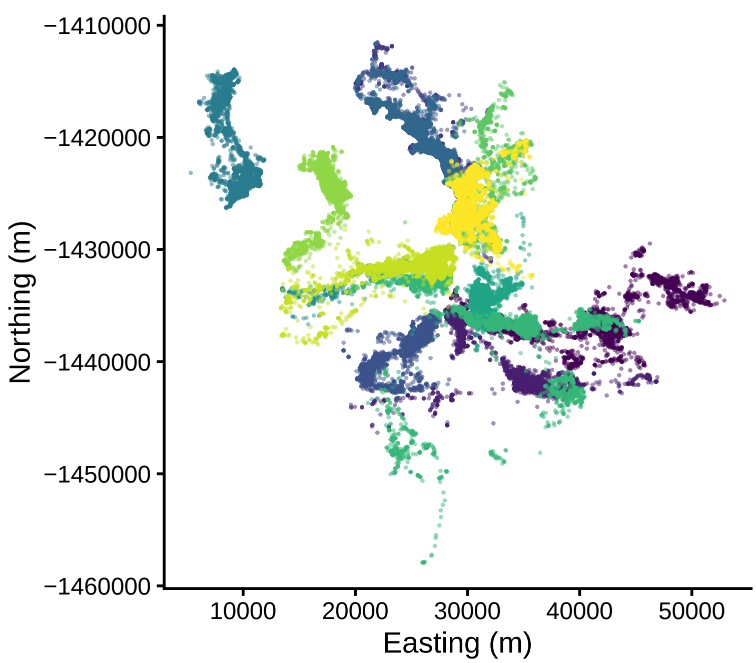
Use the `amt` package to create a trajectory object from the cleaned data.

```r
buffalo_all <- buffalo_clean %>% mk_track(id = id,
                                          lon,
                                          lat,
                                          time,
                                          all_cols = T,
                                          crs = 4326) %>%
  # Transformation to GDA94 / Geoscience Australia Lambert (https://epsg.io/3112)
  transform_coords(crs_to = 3112, crs_from = 4326) %>% arrange(id)

# Number of IDs and locations
buffalo_all %>%
  summarise(n_ids = n_distinct(id),
            n_locations = n())
```

```
# A tibble: 1 x 2
  n_ids n_locations
  <int>       <int>
1    13      107554
```

Plot the data coloured by time

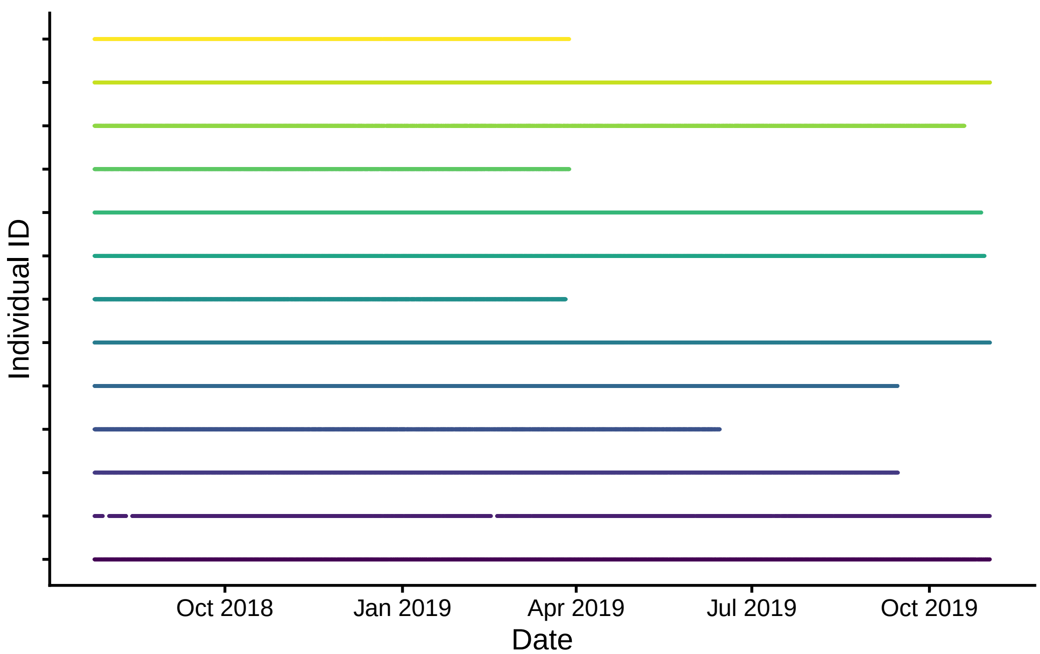
```
buffalo_all %>%
  ggplot(aes(x = x_, y = y_, colour = id)) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "none")
```




```
ggsave("outputs/data_prep/buffalo_djelk_map.png",
       width = 150, height = 150, units = "mm",  dpi = 1000)
```

Plot a timeline of each individual

```r
buffalo_all %>%
  ggplot(aes(x = t_, y = as.factor(id), colour = as.factor(id))) +
  geom_point(alpha = 0.5, size = 0.01) +
  scale_x_datetime("Date") +
  scale_y_discrete("Individual ID") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "none",
        axis.text.y = element_blank())
```

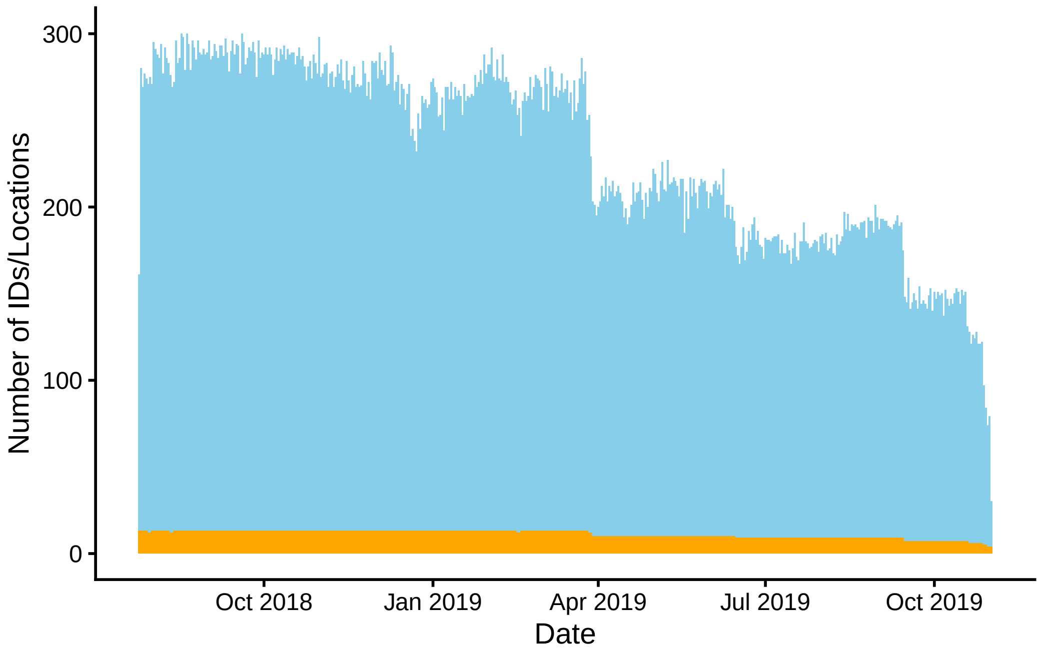



```r
ggsave("outputs/data_prep/buffalo_djelk_timeline.png",
       width = 150, height = 150, units = "mm",  dpi = 600)
```

Number of location per day

```r
buffalo_daily_locations <- buffalo_all %>%
  mutate(date = lubridate::date(t_)) %>%
  dplyr::group_by(date) %>%
  summarise(
    n_locations = n(),
    n_individuals = n_distinct(id)
)
```

```r
buffalo_daily_locations %>%
  ggplot() +
    geom_col(aes(x = date, y = n_locations),
             fill = "skyblue") +
    geom_col(aes(x = date, y = n_individuals),
             fill = "orange") +
    scale_x_date("Date") +
    scale_y_continuous("Number of IDs/Locations") +
    scale_colour_viridis_d() +
    theme_classic() +
    theme(legend.position = "none")
```




```r
ggsave("outputs/data_prep/buffalo_djelk_daily_locations.png",
       width = 150, height = 80, units = "mm",  dpi = 600)
```

Generating the data to fit a deepSSF model

Create a steps object to check the step lengths

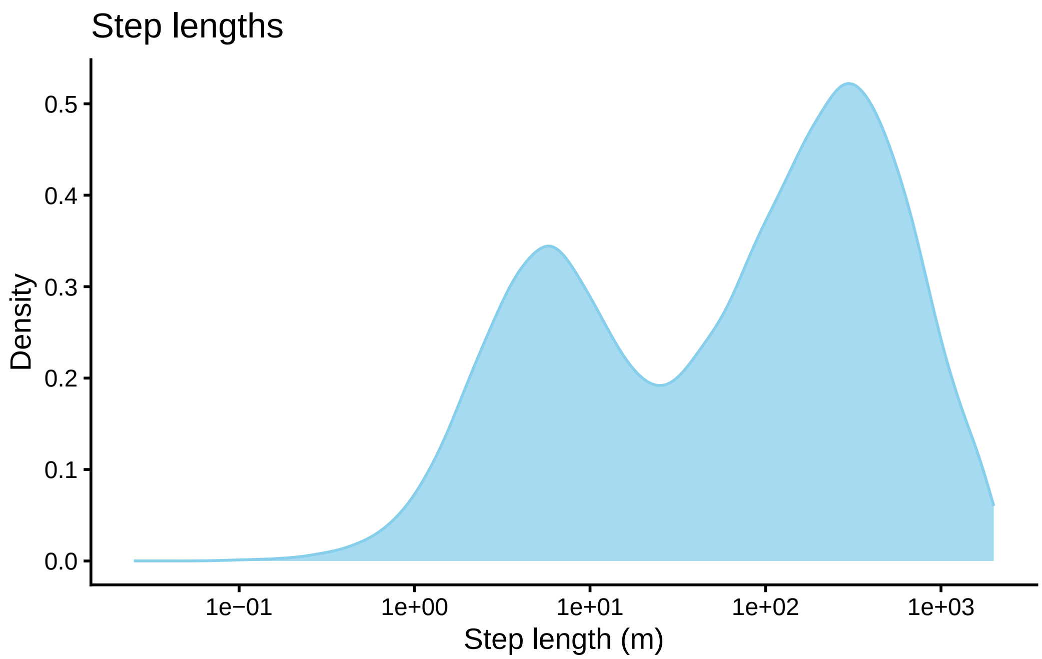
```r
# nest the data by individual
buffalo_all_nested <- buffalo_all %>% arrange(id) %>% nest(data = -"id")
```

```r
buffalo_all_nested_steps <- buffalo_all_nested %>%
  mutate(steps = map(data, function(x)
    x %>%
      # track_resample(rate = hours(1), tolerance = minutes(10)) %>%
      steps()))

# unnest the data after creating 'steps' objects
buffalo_all_steps <- buffalo_all_nested_steps %>%
  amt::select(id, steps) %>%
  amt::unnest(cols = steps)

head(buffalo_all_steps)
```

```
# A tibble: 6 x 11
  id       x1_    x2_       y1_       y2_   sl_ direction_p     ta_
* <chr>  <dbl>  <dbl>     <dbl>     <dbl> <dbl>       <dbl>   <dbl>
1 2005  41941. 41969. -1435875. -1435671. 206.         1.43  NA
2 2005  41969. 41922. -1435671. -1435654.  50.7        2.80   1.37
3 2005  41922. 41779. -1435654. -1435601. 152.         2.78  -0.0214
4 2005  41779. 41841. -1435601. -1435635.  70.7       -0.507  2.99
5 2005  41841. 41655. -1435635. -1435604. 188.         2.98  -2.80
6 2005  41655. 41619. -1435604. -1435608.  37.1       -3.02   0.285
# i 3 more variables: t1_ <dttm>, t2_ <dttm>, dt_ <drtn>
```

Plot step lengths and set the buffer distance

```r
# plot step lengths
buffalo_all_steps %>%
  filter(sl_ < 2000) %>%
  ggplot() +
  geom_density(aes(x = sl_), #, bins = 50
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  # scale_x_continuous("Step length (m)") + #, limits = c(-25, 1250)
  scale_x_log10("Step length (m)") + #, limits = c(-25, 1250)
  scale_y_continuous("Density") +
  ggtitle("Step lengths") +
  theme_classic()
```




```r
# proportion of steps less than the buffer distance
# the distance should be a multiple of the resolution
distance <- 1250
res <- 25

# check if multiple of the resolution?
distance %% res == 0
```

```
[1] TRUE
```

```r
# buffer distance (add the resolution of the cell/2 as we want a centre cell)
buffer <- distance + (res/2)

# calculate the number of cells in each axis
nxn_cells <- buffer*2/res
print(paste0("Number of cells in each axis: ", nxn_cells))
```

```
[1] "Number of cells in each axis: 101"
```

```r
# calculate proportion
prop_steps <- buffalo_all_steps %>% filter(x2_ - x1_ > -buffer &
                                           x2_ - x1_ < buffer &
                                           y2_ - y1_ > -buffer &
                                           y2_ - y1_ < buffer) %>%
```

```
  nrow() / buffalo_all_steps %>% nrow()

print(paste0("Proportion of steps less than ", buffer, "m in x or y direction: ", round(prop
```

```
[1] "Proportion of steps less than 1262.5m in x or y direction: 0.9637"
```

```
# maxmimum distance at corner
max_dist <- sqrt(buffer^2 + buffer^2)
print(paste0("Maximum distance at corner: ", round(max_dist, 2)))
```

```
[1] "Maximum distance at corner: 1785.44"
```

Prepare and export data for single layer saving (Python script)

We also have another script that saves all local layers as individual npy arrays, which works better when there are many steps per id (to prevent large tif files), and can make things a bit easier for training models with many ids.

```
buffalo_all_steps_save <- buffalo_all_steps %>% mutate(

  dt_minute = as.numeric(round(difftime(t2_, t1_, units = "mins"), 0)),
  dt_hour = as.numeric(round(difftime(t2_, t1_, units = "hours"), 2)),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t1 = round(lubridate::hour(t1_) + lubridate::minute(t1_)/60,2),
  yday_t1 = round(lubridate::yday(t1_) + lubridate::hour(t1_)/24,2),

  hour_t1_sin1 = sin(2*pi*hour_t1/24),
  hour_t1_cos1 = cos(2*pi*hour_t1/24),
  hour_t1_sin2 = sin(4*pi*hour_t1/24),
  hour_t1_cos2 = cos(4*pi*hour_t1/24),

  yday_t1_sin1 = sin(2*pi*yday_t1/365.25),
  yday_t1_cos1 = cos(2*pi*yday_t1/365.25),
  yday_t1_sin2 = sin(4*pi*yday_t1/365.25),
  yday_t1_cos2 = cos(4*pi*yday_t1/365.25),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t2 = round(lubridate::hour(t2_) + lubridate::minute(t2_)/60,2),
  yday_t2 = round(lubridate::yday(t2_) + lubridate::hour(t2_)/24,2),

  bearing = direction_p,
```

```
  bearing_sin = sin(bearing),
  bearing_cos = cos(bearing)

)

buffalo_all_steps_save
```

```
# A tibble: 107,541 x 28
   id       x1_    x2_       y1_       y2_    sl_ direction_p     ta_
 * <chr>  <dbl>  <dbl>     <dbl>     <dbl>  <dbl>       <dbl>   <dbl>
 1 2005  41941. 41969. -1435875. -1435671. 206.          1.43  NA
 2 2005  41969. 41922. -1435671. -1435654.  50.7         2.80   1.37
 3 2005  41922. 41779. -1435654. -1435601. 152.          2.78  -0.0214
 4 2005  41779. 41841. -1435601. -1435635.  70.7        -0.507  2.99
 5 2005  41841. 41655. -1435635. -1435604. 188.          2.98  -2.80
 6 2005  41655. 41619. -1435604. -1435608.  37.1        -3.02   0.285
 7 2005  41619. 41688. -1435608. -1436126. 522.         -1.44   1.58
 8 2005  41688. 41684. -1436126. -1436119.   8.13        2.10  -2.75
 9 2005  41684. 41675. -1436119. -1436119.   9.75       -3.13   1.05
10 2005  41675. 41425. -1436119. -1435938. 308.          2.52  -0.635
# i 107,531 more rows
# i 20 more variables: t1_ <dttm>, t2_ <dttm>, dt_ <drtn>, dt_minute <dbl>,
#   dt_hour <dbl>, hour_t1 <dbl>, yday_t1 <dbl>, hour_t1_sin1 <dbl>,
#   hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>, hour_t1_cos2 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, hour_t2 <dbl>, yday_t2 <dbl>, bearing <dbl>,
#   bearing_sin <dbl>, bearing_cos <dbl>
```

Save the data to a csv file

```
# remove steps that fall outside of the local spatial extent
buffalo_all_steps_save <- buffalo_all_steps_save %>%
  filter(x2_ - x1_ > -distance &
           x2_ - x1_ < distance &
           y2_ - y1_ > -distance &
           y2_ - y1_ < distance) %>%
  drop_na(ta_)

buffalo_all_steps_save
```

```
# A tibble: 103,558 x 28
   id       x1_    x2_       y1_       y2_    sl_ direction_p     ta_
```

```
   <chr>  <dbl>  <dbl>     <dbl>     <dbl>  <dbl>       <dbl>   <dbl>
 1 2005  41969. 41922. -1435671. -1435654.  50.7        2.80   1.37  
 2 2005  41922. 41779. -1435654. -1435601. 152.         2.78  -0.0214
 3 2005  41779. 41841. -1435601. -1435635.  70.7       -0.507  2.99  
 4 2005  41841. 41655. -1435635. -1435604. 188.         2.98  -2.80  
 5 2005  41655. 41619. -1435604. -1435608.  37.1       -3.02   0.285 
 6 2005  41619. 41688. -1435608. -1436126. 522.        -1.44   1.58  
 7 2005  41688. 41684. -1436126. -1436119.   8.13       2.10  -2.75  
 8 2005  41684. 41675. -1436119. -1436119.   9.75      -3.13   1.05  
 9 2005  41675. 41425. -1436119. -1435938. 308.         2.52  -0.635 
10 2005  41425. 40563. -1435938. -1435841. 867.         3.03   0.513 
# i 103,548 more rows
# i 20 more variables: t1_ <dttm>, t2_ <dttm>, dt_ <drtn>, dt_minute <dbl>,
#   dt_hour <dbl>, hour_t1 <dbl>, yday_t1 <dbl>, hour_t1_sin1 <dbl>,
#   hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>, hour_t1_cos2 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, hour_t2 <dbl>, yday_t2 <dbl>, bearing <dbl>,
#   bearing_sin <dbl>, bearing_cos <dbl>
```

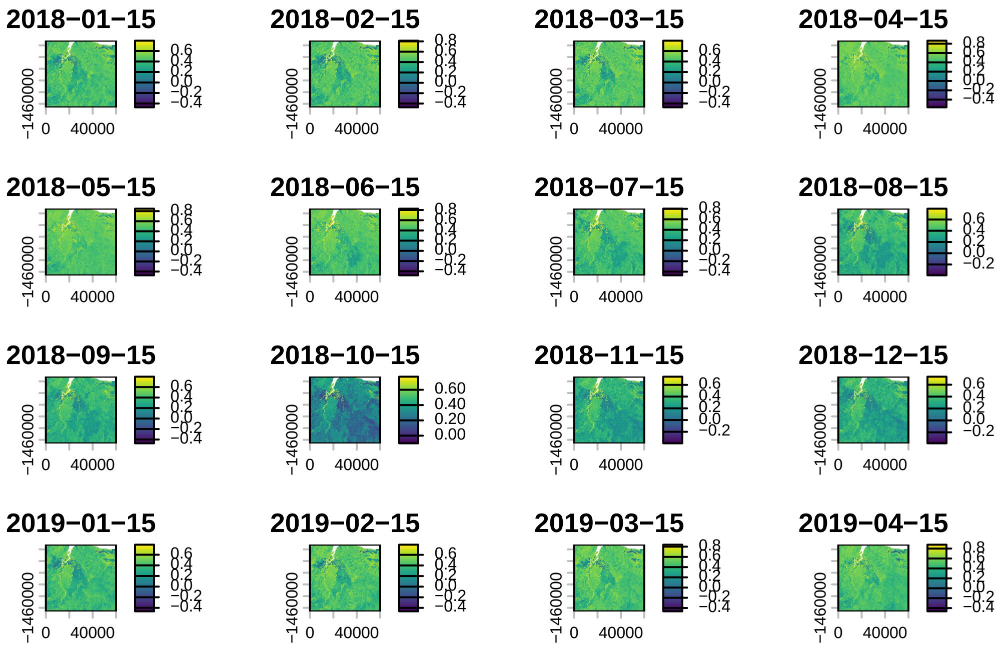
```
buffalo_all_steps_save %>% write_csv(paste0("data/buffalo_djelk_all_steps_nxn", nxn_cells,".
```

In our case over 96% of the steps were less than a distance of 1262.5m in either the x or y direction, which would mean that including 101 x 101 cells (which would be 2525 m x 2525 m) would include at least 96% of the steps. As the distance is further to the corner (but still less than the buffer distance in the x and y directions), there will be step lengths longer than the buffer distance (up to 1785 m).

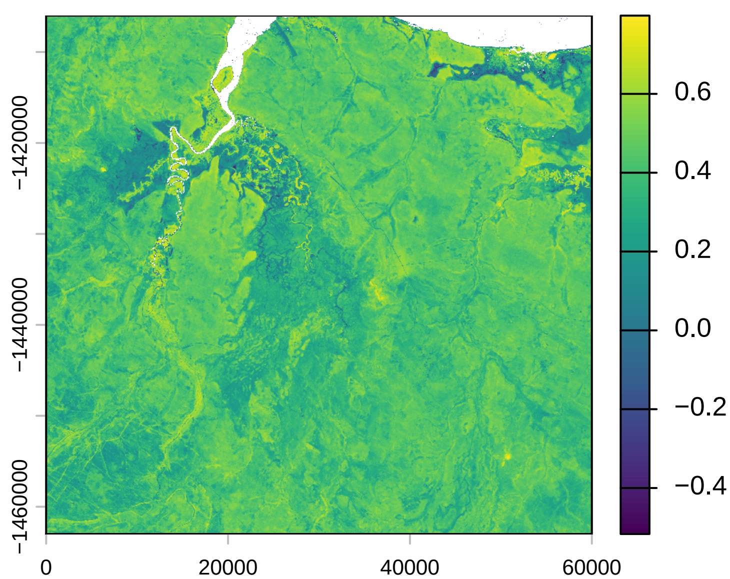
Read in the environmental covariates

```
ndvi_projected <- rast("mapping/cropped rasters/ndvi_GEE_projected_watermask20230207.tif")
terra::time(ndvi_projected) <- as.POSIXct(lubridate::ymd("2018-01-15") + months(0:23))
slope <- rast("mapping/cropped rasters/slope_raster.tif")
veg_herby <- rast("mapping/cropped rasters/veg_herby.tif")
canopy_cover <- rast("mapping/cropped rasters/canopy_cover.tif")

# change the names (these will become the column names when extracting
# covariate values at the used and random steps)
names(ndvi_projected) <- rep("ndvi", terra::nlyr(ndvi_projected))
names(slope) <- "slope"
names(veg_herby) <- "veg_herby"
names(canopy_cover) <- "canopy_cover"
```

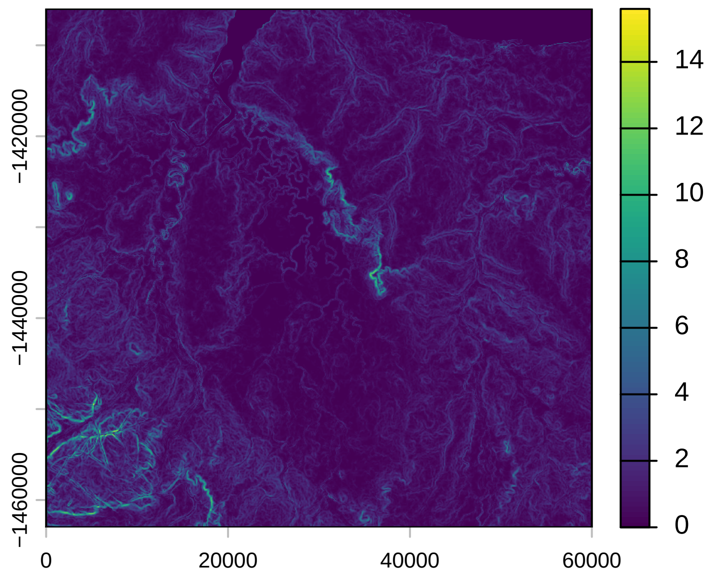
```r
# to plot the rasters
plot(ndvi_projected)
```



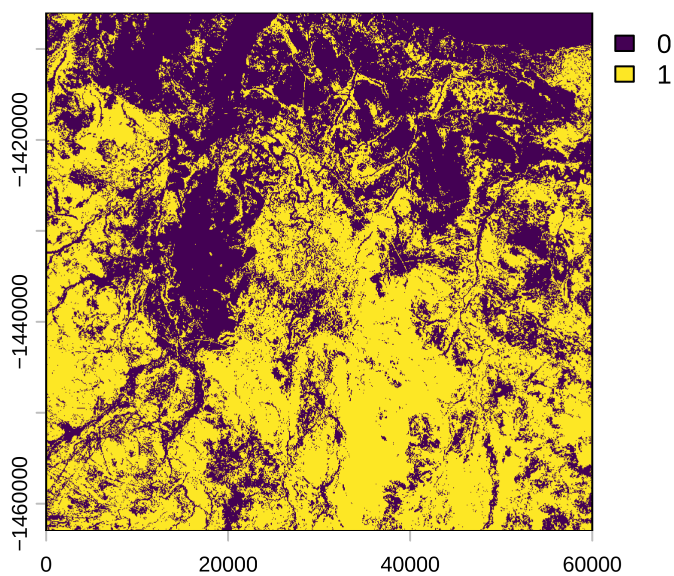

```r
plot(ndvi_projected[[1]])
```



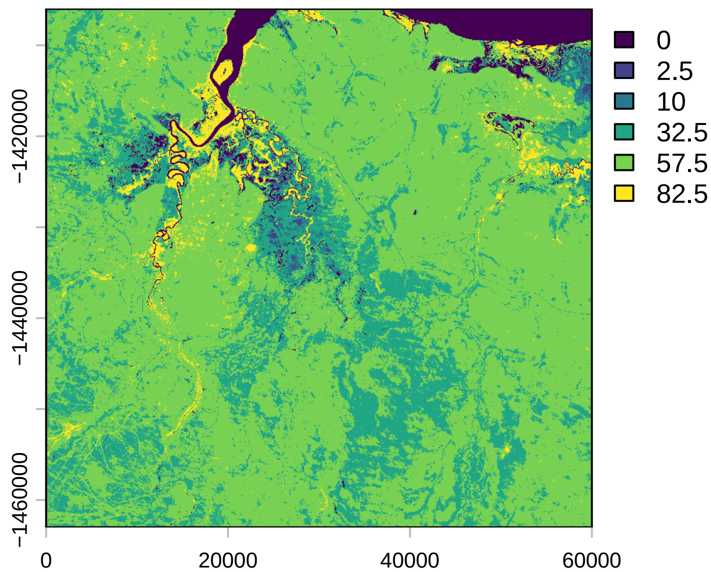

```r
plot(slope)
```




```r
plot(veg_herby)
```




```r
plot(canopy_cover)
```

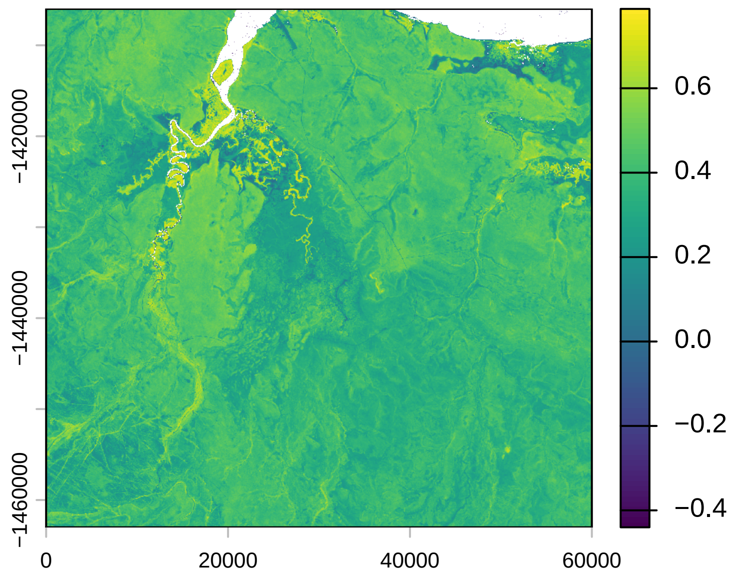



```
# get the resolution from the covariates
res <- terra::res(ndvi_projected)[1]
```

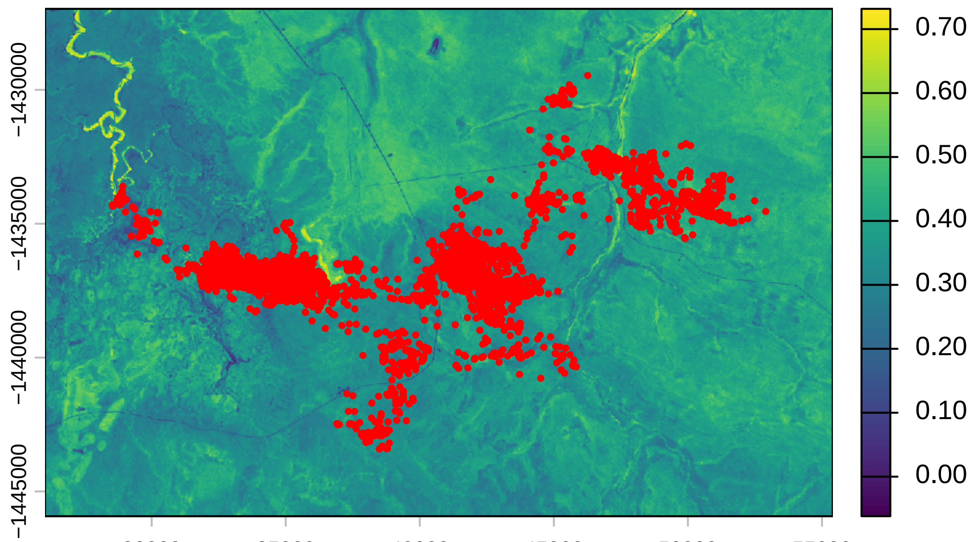
Create some layers for testing the deepSSF package

These will be cropped to the extent of the ID 2005 data. We will also take an average of the NDVI values for simplicity.

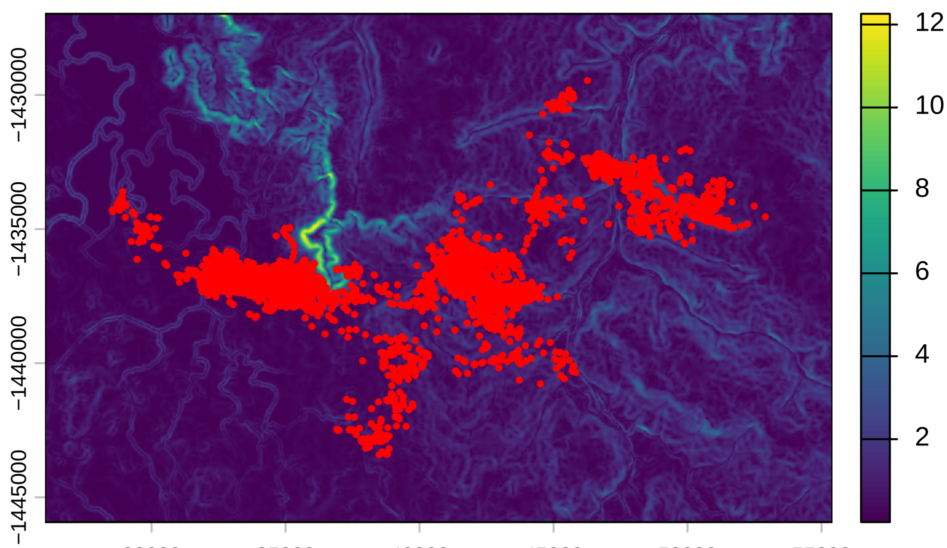
```
buffer_distance <- 2500 # add a generous buffer distance to ensure we capture all steps for

ndvi_average <- terra::app(ndvi_projected, mean)
plot(ndvi_average)
```




```
# create a buffer around the buffalo 2005 data
buffalo_clean_2005_sf <- buffalo_clean_2005 %>%
  st_as_sf(coords = c("lon", "lat"), crs = 4326) %>%
  st_transform(crs = 3112)

buffalo_clean_2005_extent <- buffalo_clean_2005_sf %>%
  st_buffer(dist = buffer_distance) %>%
  st_bbox() %>%
  st_as_sfc()

# crop the rasters to the extent of the buffalo 2005 data with the buffer distance
ndvi_2005 <- terra::crop(ndvi_average, buffalo_clean_2005_extent)
slope_2005 <- terra::crop(slope, buffalo_clean_2005_extent)

plot(ndvi_2005)
points(buffalo_clean_2005_sf$geometry, pch = 16, cex = 0.5, col = "red")
```




```
plot(slope_2005)
points(buffalo_clean_2005_sf$geometry, pch = 16, cex = 0.5, col = "red")
```




```
# write the rasters to file
terra::writeRaster(ndvi_2005, "mapping/cropped rasters/ndvi_2005.tif", overwrite = T)
terra::writeRaster(slope_2005, "mapping/cropped rasters/slope_2005.tif", overwrite = T)
```

Set up the spatial extent of the local covariates

We calculate the number of cells in each axis, and set the lag between locations. Typically this will just be 1, as we want the next-step to be the target. However, it may be possible to improve model training by using different lags and including the time difference between locations as a covariate. I expect that this would help to predict across different time scales and not be restricted to the time scale that the data was collected at.

```
# calculate the number of cells in each axis
nxn_cells <- buffer*2/res
print(paste0("Number of cells in each axis: ", nxn_cells))
```

```
[1] "Number of cells in each axis: 101"
```

```
# hourly lag - to set larger time differences between locations
hourly_lag <- 1
```

Loop over each individual and save the local rasters

This is the main function of the script. It loops over each individual and calculates step level information, such as the location and time of the next step (x2, y2, t2), step length, turning angle, and different time components.

It also determines where to crop the local layers by subtracting and adding the buffer distance from the location of each step. It then crops out the local layers for each step and saves them as a list entry into the data frame.

It then creates a raster stack (with the number of layers equal to the number of steps) for each covariate and saves it as a tif.

For each individual, this will therefore create a csv containing the step level information, and a tif for each covariate containing the local layers for each step.

It takes quite a while to run, around 1 hour per individual in our case, so we will illustrate the function with 10 steps per individual. Uncomment the line specified in the loop to use the full dataset.

```
# for(i in 1:length(buffalo_ids)) {
for(i in 1:1) { # select just the first ID

  buffalo_data <- buffalo_all %>% filter(id == buffalo_ids[i])
  # all data for that individual
  buffalo_data <- buffalo_data %>% arrange(t_)

  # to use a subset of the data for that individual for testing
  # COMMENT THIS LINE OUT TO USE THE FULL DATASET
  buffalo_data <- buffalo_data %>% arrange(t_) |>  slice(1:100)
```

```r
n_samples <- nrow(buffalo_data)

tic()

buffalo_data_covs <- buffalo_data %>% mutate(

  x1_ = x_,
  y1_ = y_,
  x2_ = lead(x1_, n = hourly_lag, default = NA),
  y2_ = lead(y1_, n = hourly_lag, default = NA),
  x2_cent = x2_ - x1_,
  y2_cent = y2_ - y1_,
  t2_ = lead(t_, n = hourly_lag, default = NA),
  t_diff = round(difftime(t2_, t_, units = "hours"),0),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t1 = round(lubridate::hour(t_) + lubridate::minute(t_)/60,2),
  yday_t1 = round(lubridate::yday(t_) + lubridate::hour(t_)/24,2),

  hour_t1_sin1 = sin(2*pi*hour_t1/24),
  hour_t1_cos1 = cos(2*pi*hour_t1/24),
  hour_t1_sin2 = sin(4*pi*hour_t1/24),
  hour_t1_cos2 = cos(4*pi*hour_t1/24),
  hour_t1_sin3 = sin(6*pi*hour_t1/24),
  hour_t1_cos3 = cos(6*pi*hour_t1/24),

  yday_t1_sin1 = sin(2*pi*yday_t1/365.25),
  yday_t1_cos1 = cos(2*pi*yday_t1/365.25),
  yday_t1_sin2 = sin(4*pi*yday_t1/365.25),
  yday_t1_cos2 = cos(4*pi*yday_t1/365.25),
  yday_t1_sin3 = sin(6*pi*yday_t1/365.25),
  yday_t1_cos3 = cos(6*pi*yday_t1/365.25),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t2 = round(lubridate::hour(t2_) + lubridate::minute(t2_)/60,2),
  hour_t2_sin = sin(2*pi*hour_t2/24),
  hour_t2_cos = cos(2*pi*hour_t2/24),

  yday_t2 = round(lubridate::yday(t2_) + lubridate::hour(t2_)/24,2),
  yday_t2_sin = sin(2*pi*yday_t2/365.25),
  yday_t2_cos = cos(2*pi*yday_t2/365.25),

  sl = c(sqrt(diff(y_)^2 + diff(x_)^2), NA),
  log_sl = log(sl),
```

```
  bearing = c(atan2(diff(y_), diff(x_)), NA),
  bearing_sin = sin(bearing),
  bearing_cos = cos(bearing),
  ta = c(NA, ifelse(
    diff(bearing) > pi, diff(bearing)-(2*pi), ifelse(
      diff(bearing) < -pi, diff(bearing)+(2*pi), diff(bearing)))),
  cos_ta = cos(ta),

  # extent for cropping the spatial covariates
  x_min = x_ - buffer,
  x_max = x_ + buffer,
  y_min = y_ - buffer,
  y_max = y_ + buffer,


  # crop out and store the local covariates centered on the animal's location
  # with an extent set in the previous chunk
) %>% rowwise() %>% mutate(

  # the extent needs to be the same for saving the stack of rasters
  extent_00centre = list(ext(x_min - x_, x_max - x_, y_min - y_, y_max - y_)),

  # NDVI
  ndvi_index = which.min(abs(difftime(t_, terra::time(ndvi_projected)))),
  ndvi_cent = list({
    ndvi_cent = crop(ndvi_projected[[ndvi_index]], ext(x_min, x_max, y_min, y_max))
    ext(ndvi_cent) <- extent_00centre
    ndvi_cent
    }),

  # herbaceous vegetation
  veg_herby_cent = list({
    veg_herby_cent = crop(veg_herby, ext(x_min, x_max, y_min, y_max))
    ext(veg_herby_cent) <- extent_00centre
    veg_herby_cent
    }),

  # canopy cover
  canopy_cover_cent = list({
    canopy_cover_cent = crop(canopy_cover, ext(x_min, x_max, y_min, y_max))
    ext(canopy_cover_cent) <- extent_00centre
    canopy_cover_cent
    }),
```

```
  # slope
  slope_cent = list({
    slope_cent <- crop(slope, ext(x_min, x_max, y_min, y_max))
    ext(slope_cent) <- extent_00centre
    slope_cent
    }),

  # presence (next-step)
  pres_template = list({
    pres_template <- crop(slope, ext(x_min, x_max, y_min, y_max))
    pres_template
  }),

  # Strict within cell method
  points_vect_cent_x2_within_cell = list(terra::vect(cbind(x2_, y2_), type = "points", crs
  pres_cent_x2_within_cell = list({
    pres_cent_x2_within_cell <- rasterize(points_vect_cent_x2_within_cell, pres_template,
    ext(pres_cent_x2_within_cell) <- extent_00centre
    pres_cent_x2_within_cell
    })

) %>% ungroup() # to remove the 'rowwise' class


toc()

# remove steps that fall outside of the local spatial extent
buffalo_data_covs <- buffalo_data_covs %>%
  filter(x2_cent > -buffer & x2_cent < buffer & y2_cent > -buffer & y2_cent < buffer) %>%
  drop_na(ta)

# remove the columns that are no longer needed
buffalo_data_df <- buffalo_data_covs %>%
dplyr::select(
  -extent_00centre,
  -ndvi_cent,
  -veg_herby_cent,
  -canopy_cover_cent,
  -slope_cent,
  -pres_template,
  -points_vect_cent_x2_within_cell,
  -pres_cent_x2_within_cell)

# save the data
```

```
  write_csv(buffalo_data_df, paste0("buffalo_local_data_id/buffalo_temporal_cont_", buffalo_
                                    "_data_df_lag_", hourly_lag, "hr_n", n_samples, ".csv"))

  # saving the raster objects
  rast(buffalo_data_covs$ndvi_cent) %>%
    writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_ndvi_cent",
                       nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
                overwrite = T)

  rast(buffalo_data_covs$veg_herby_cent) %>%
    writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_herby_cent",
                       nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
                overwrite = T)

  rast(buffalo_data_covs$canopy_cover_cent) %>%
    writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_canopy_cent",
                       nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
                overwrite = T)

  rast(buffalo_data_covs$slope_cent) %>%
    writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_slope_cent",
                       nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
                overwrite = T)

  rast(buffalo_data_covs$pres_cent_x2_within_cell) %>%
    writeRaster(paste0("buffalo_local_layers_id/within_cell_buffalo_", buffalo_ids[i], "_pre
                       nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
                overwrite = T)

}
```

```
6.304 sec elapsed
```

Check the outputs

```
head(buffalo_data_df)
```

```
# A tibble: 6 x 44
      x_      y_ t_                  id       x1_      y1_    x2_      y2_ x2_cent
   <dbl>   <dbl> <dttm>              <chr>  <dbl>    <dbl>  <dbl>    <dbl>   <dbl>
1 41969. -1.44e6 2018-07-25 11:04:23 2005  41969. -1.44e6 41922. -1.44e6   -
47.8
```

```
2 41922. -1.44e6 2018-07-25 12:04:39 2005  41922. -1.44e6 41779. -1.44e6  -
142.
3 41779. -1.44e6 2018-07-25 13:04:17 2005  41779. -1.44e6 41841. -1.44e6   61.8
4 41841. -1.44e6 2018-07-25 14:04:39 2005  41841. -1.44e6 41655. -1.44e6  -
186.
5 41655. -1.44e6 2018-07-25 15:04:27 2005  41655. -1.44e6 41619. -1.44e6   -
36.8
6 41619. -1.44e6 2018-07-25 16:04:24 2005  41619. -1.44e6 41688. -1.44e6   69.8
# i 35 more variables: y2_cent <dbl>, t2_ <dttm>, t_diff <drtn>, hour_t1 <dbl>,
#   yday_t1 <dbl>, hour_t1_sin1 <dbl>, hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>,
#   hour_t1_cos2 <dbl>, hour_t1_sin3 <dbl>, hour_t1_cos3 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, yday_t1_sin3 <dbl>, yday_t1_cos3 <dbl>, hour_t2 <dbl>,
#   hour_t2_sin <dbl>, hour_t2_cos <dbl>, yday_t2 <dbl>, yday_t2_sin <dbl>,
#   yday_t2_cos <dbl>, sl <dbl>, log_sl <dbl>, bearing <dbl>, ...
```

```r
head(buffalo_data_covs)
```

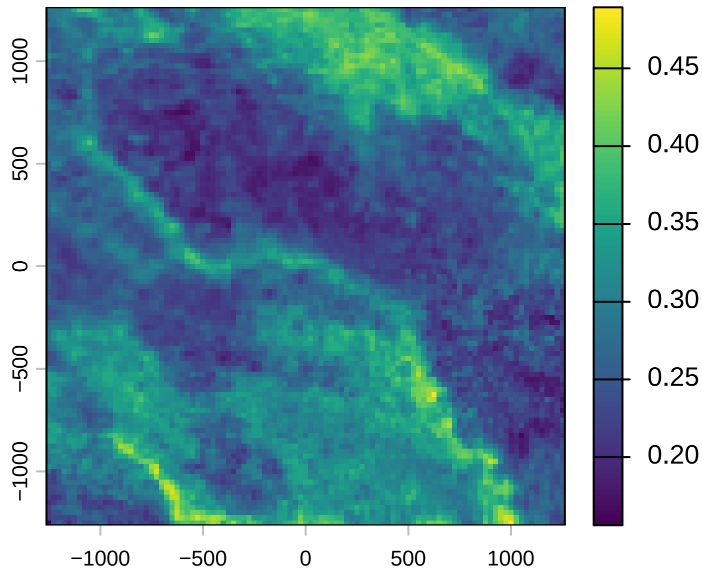
```
# A tibble: 6 x 52
      x_      y_ t_                  id       x1_     y1_    x2_     y2_ x2_cent
   <dbl>   <dbl> <dttm>              <chr>  <dbl>   <dbl>  <dbl>   <dbl>   <dbl>
1 41969. -1.44e6 2018-07-25 11:04:23 2005  41969. -1.44e6 41922. -1.44e6   -
47.8
2 41922. -1.44e6 2018-07-25 12:04:39 2005  41922. -1.44e6 41779. -1.44e6  -
142.
3 41779. -1.44e6 2018-07-25 13:04:17 2005  41779. -1.44e6 41841. -1.44e6   61.8
4 41841. -1.44e6 2018-07-25 14:04:39 2005  41841. -1.44e6 41655. -1.44e6  -
186.
5 41655. -1.44e6 2018-07-25 15:04:27 2005  41655. -1.44e6 41619. -1.44e6   -
36.8
6 41619. -1.44e6 2018-07-25 16:04:24 2005  41619. -1.44e6 41688. -1.44e6   69.8
# i 43 more variables: y2_cent <dbl>, t2_ <dttm>, t_diff <drtn>, hour_t1 <dbl>,
#   yday_t1 <dbl>, hour_t1_sin1 <dbl>, hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>,
#   hour_t1_cos2 <dbl>, hour_t1_sin3 <dbl>, hour_t1_cos3 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, yday_t1_sin3 <dbl>, yday_t1_cos3 <dbl>, hour_t2 <dbl>,
#   hour_t2_sin <dbl>, hour_t2_cos <dbl>, yday_t2 <dbl>, yday_t2_sin <dbl>,
#   yday_t2_cos <dbl>, sl <dbl>, log_sl <dbl>, bearing <dbl>, ...
```

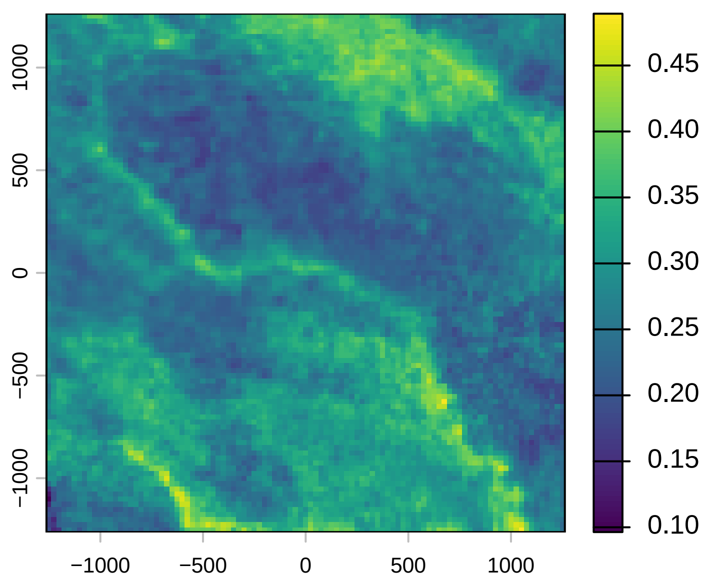
```r
which(is.na(buffalo_data_covs$ta))
```

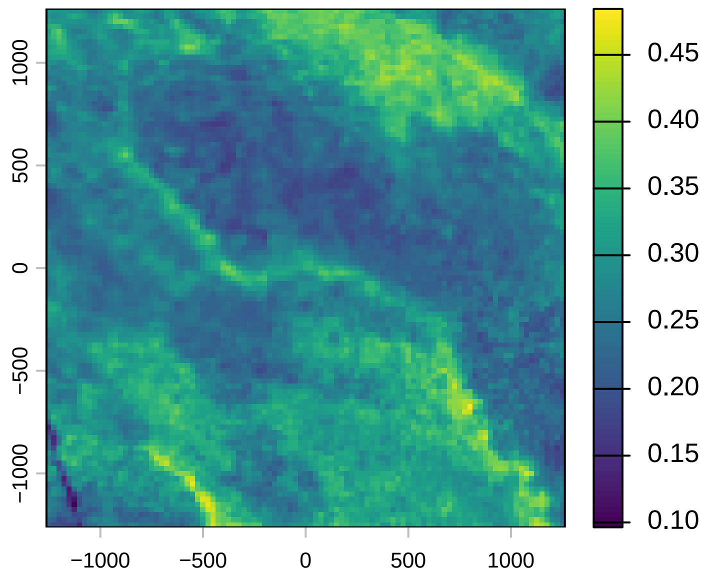
```
integer(0)
```

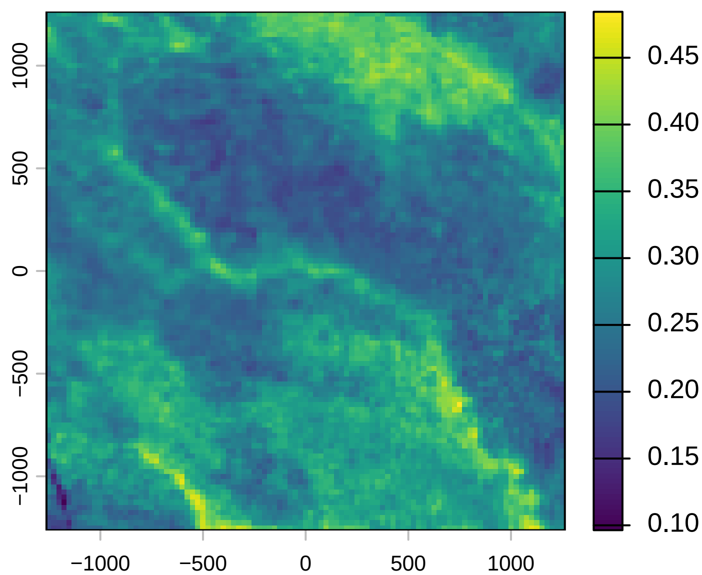
Plot a subset of the local covariates

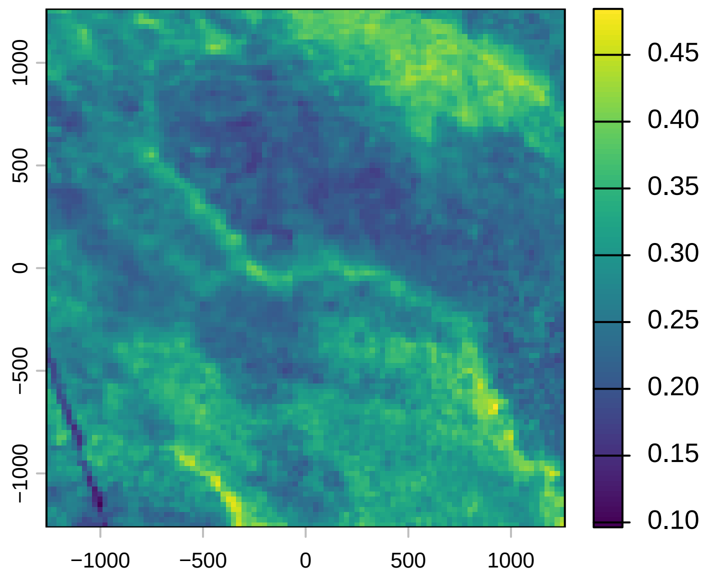
```
n_plots <- 5

# NDVI
walk(buffalo_data_covs$ndvi_cent[1:n_plots], terra::plot)
```



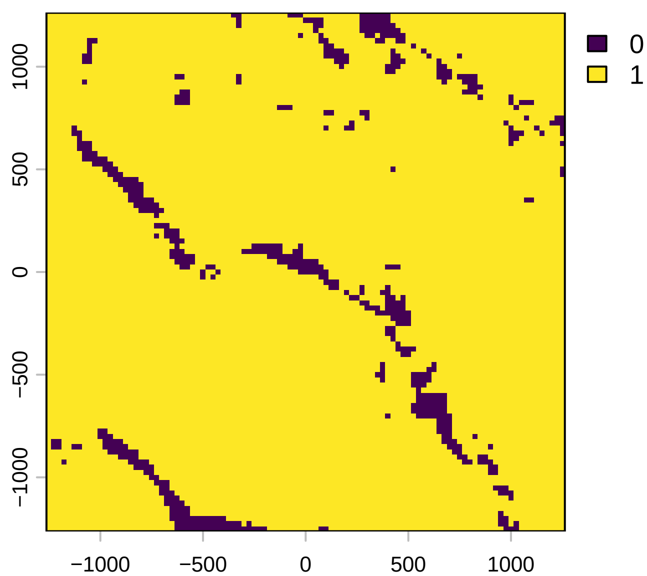


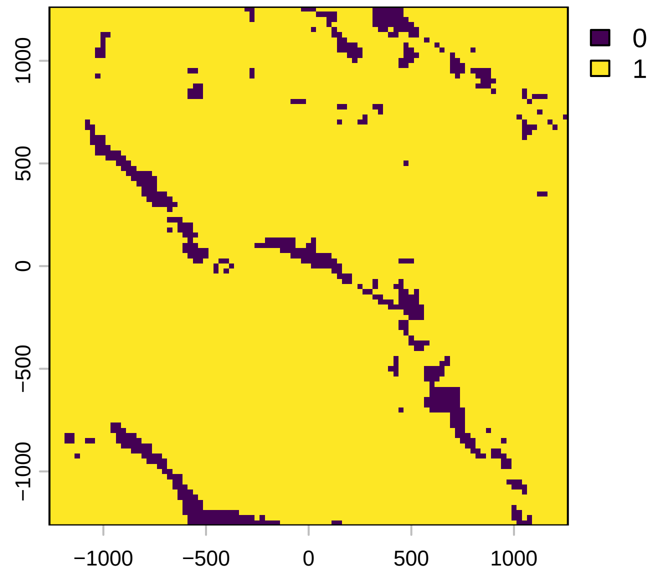

1000
500
0
−500
−1000
−1000
−500
0
500
1000
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10

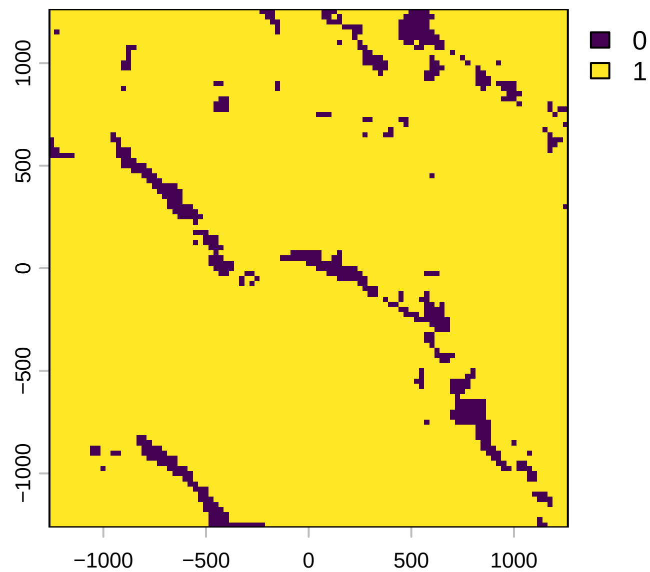

1000
500
0
−500
−1000
−1000
−500
0
500
1000
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10

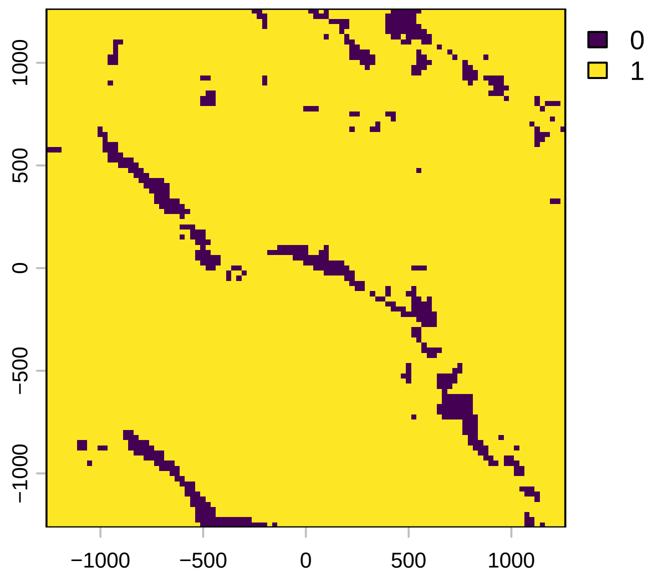


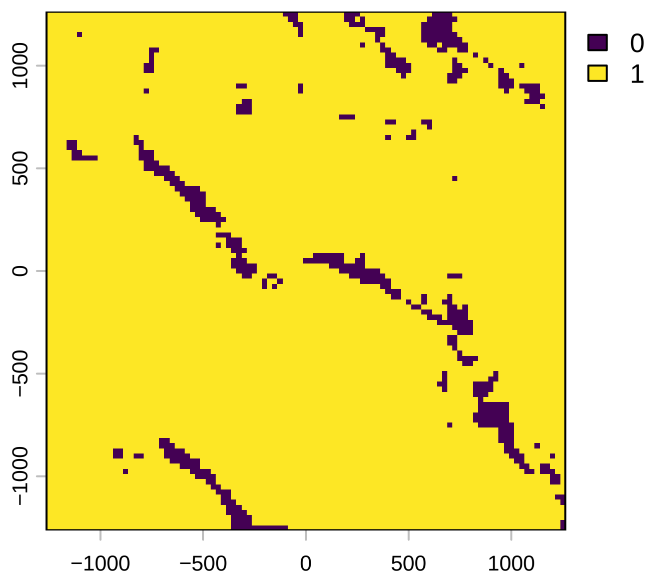

```
# Herbaceous vegetation
walk(buffalo_data_covs$veg_herby_cent[1:n_plots], terra::plot)
```



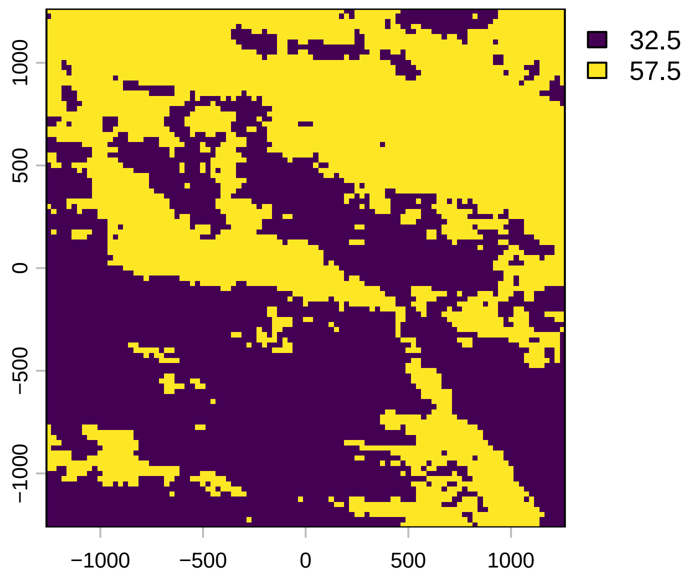

0
1
−1000
−500
0
500
1000

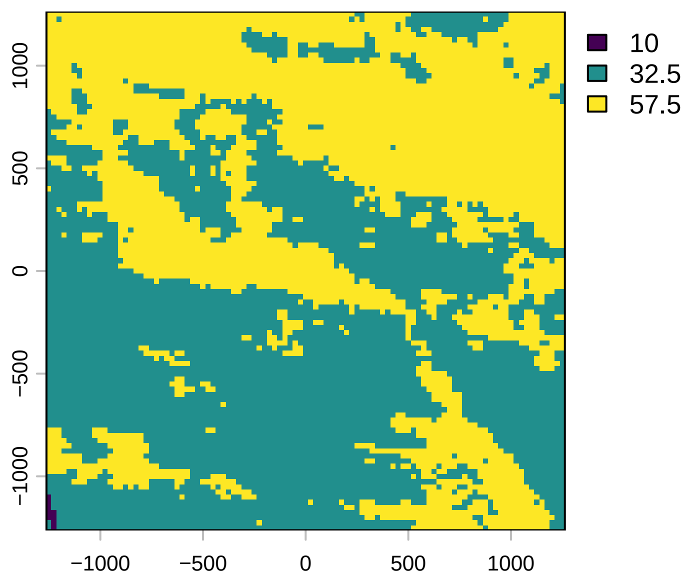

0
1
−1000
−500
0
500
1000

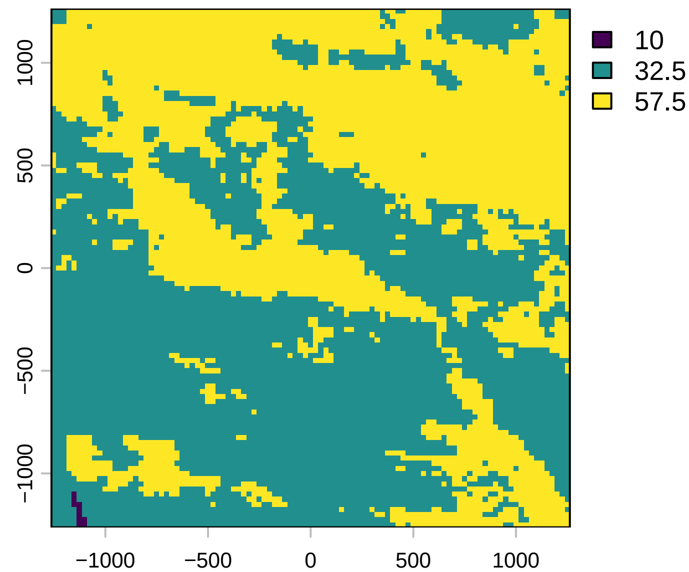


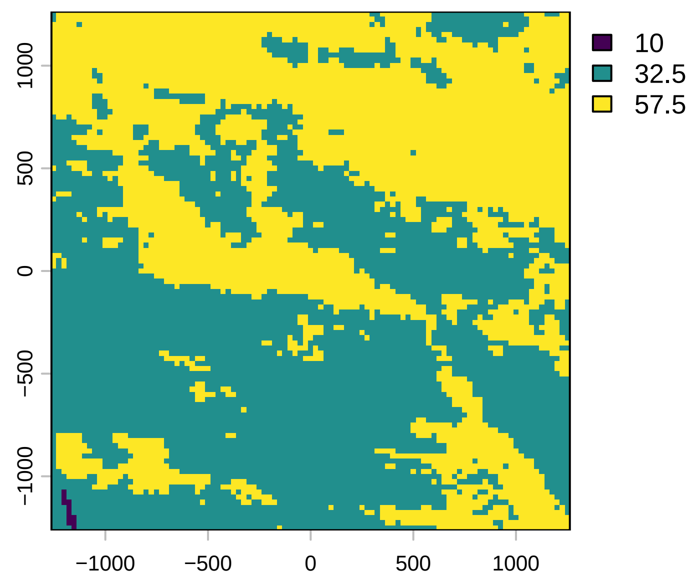


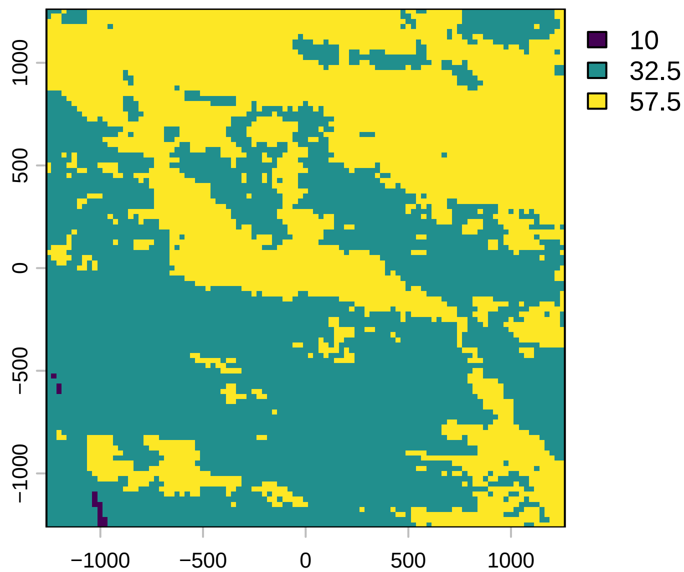

```
# Canopy cover
walk(buffalo_data_covs$canopy_cover_cent[1:n_plots], terra::plot)
```


32.5
57.5
1000
500
0
−500
−1000
−1000
−500
0
500
1000

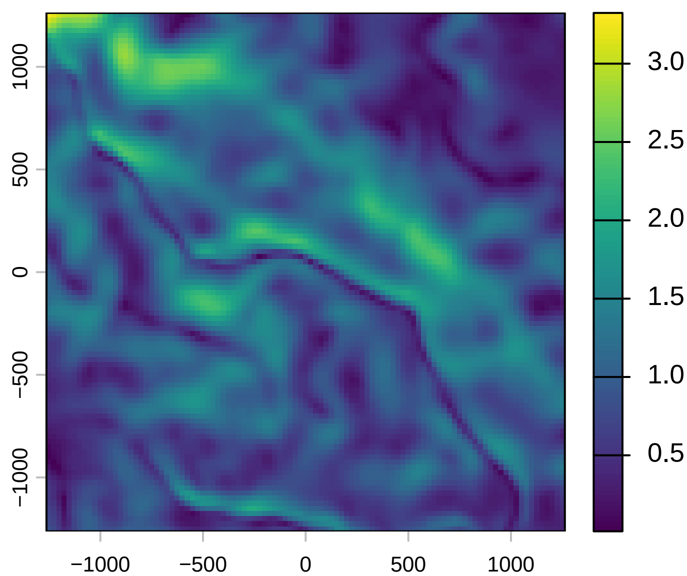

10
32.5
57.5
1000
500
0
−500
−1000
−1000
−500
0
500
1000

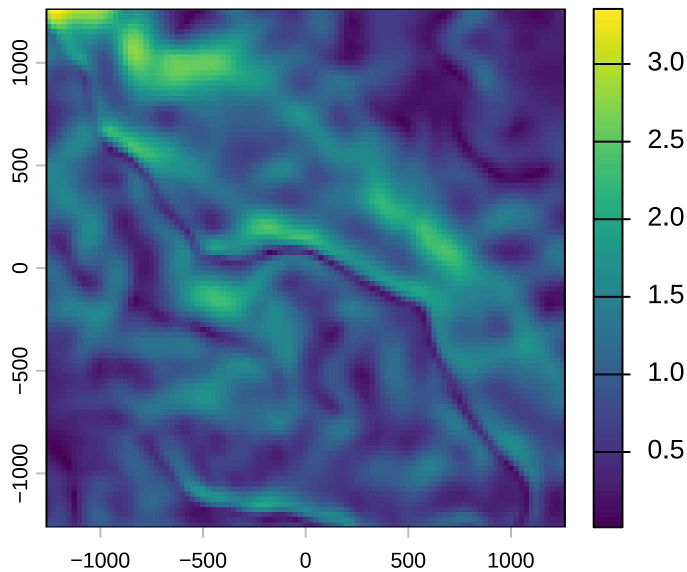

1000
500
0
−500
−1000
−1000
−500
0
500
1000
10
32.5
57.5

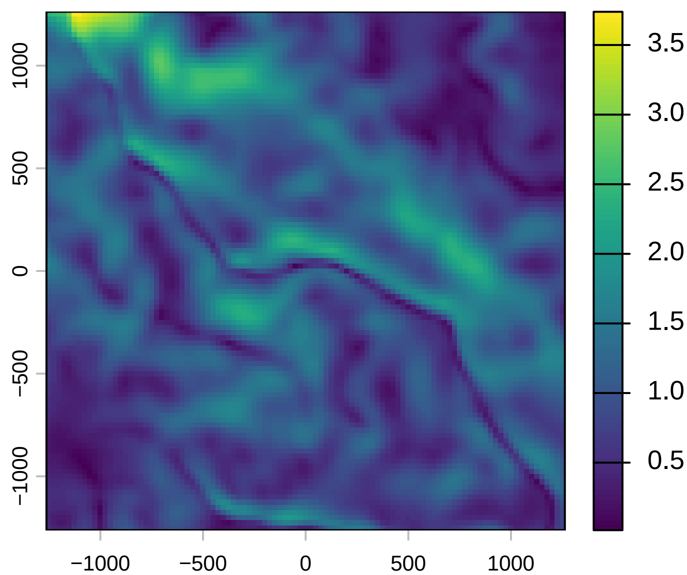

1000
500
0
−500
−1000
−1000
−500
0
500
1000
10
32.5
57.5

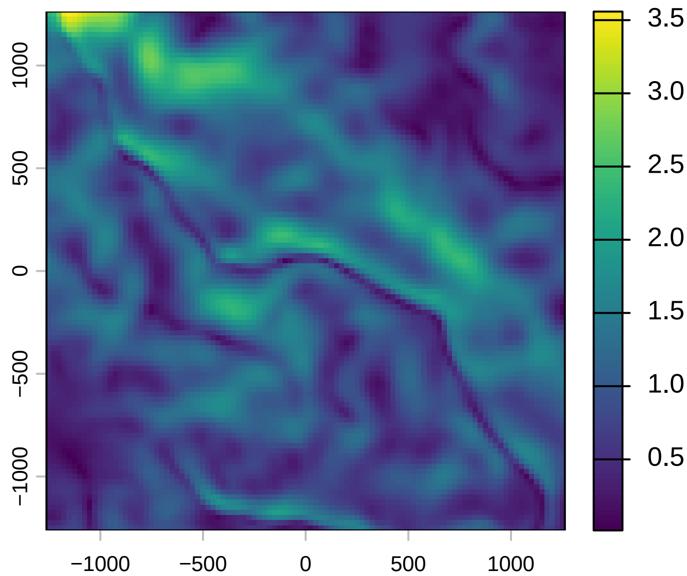


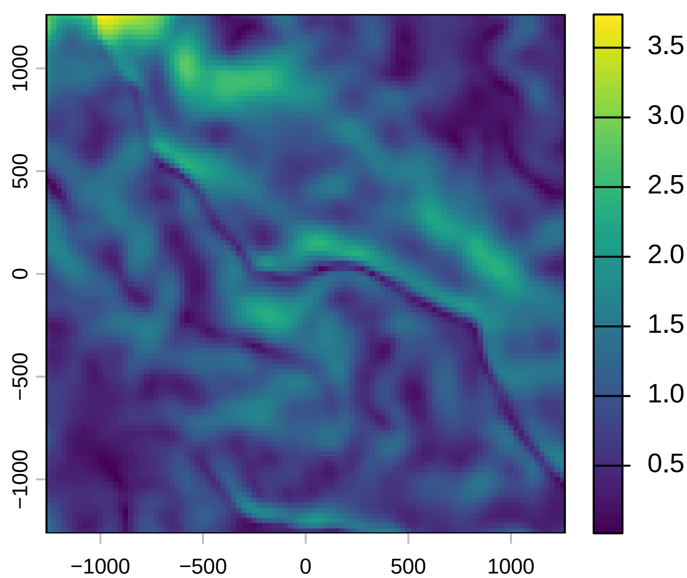

```r
# Slope
walk(buffalo_data_covs$slope_cent[1:n_plots], terra::plot)
```




1000
500
0
−500
−1000
−1000
−500
0
500
1000
3.0
2.5
2.0
1.5
1.0
0.5


1000
500
0
−500
−1000
−1000
−500
0
500
1000
3.5
3.0
2.5
2.0
1.5
1.0
0.5

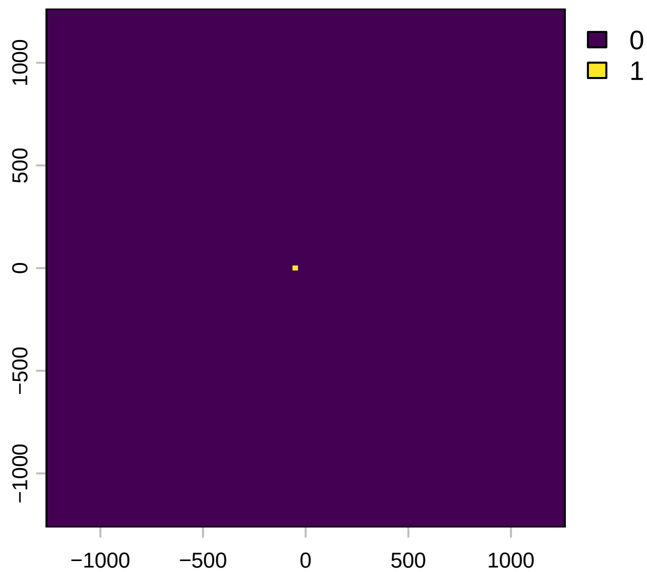


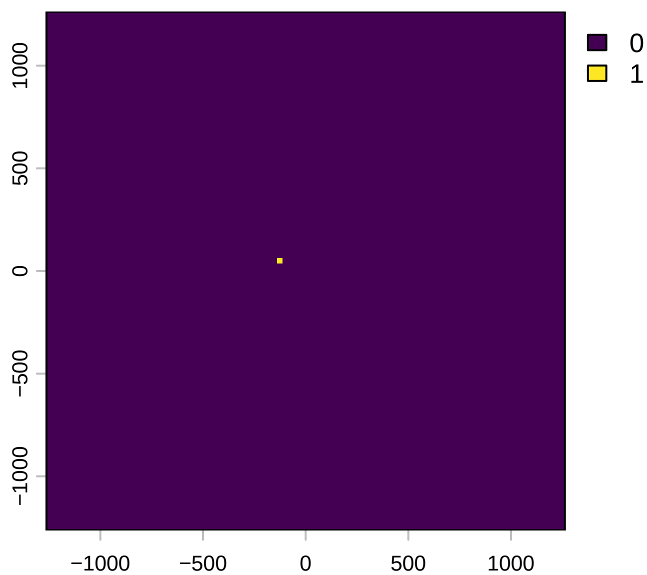


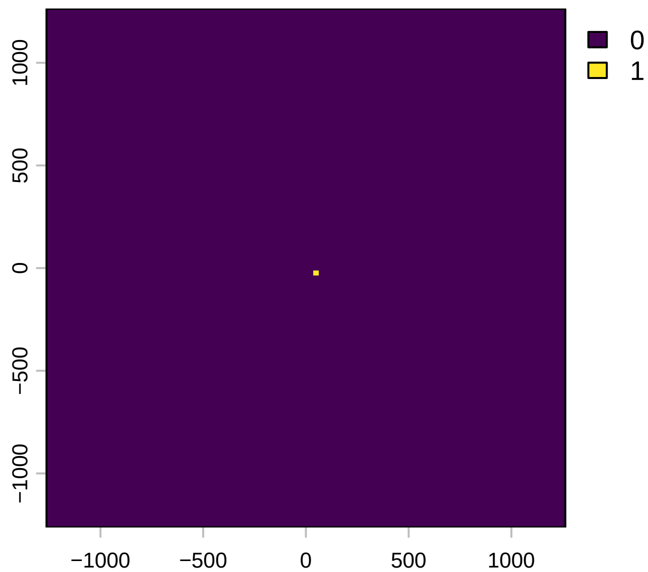

```
# Target (location of the next step)
walk(buffalo_data_covs$pres_cent_x1[1:n_plots], terra::plot)
```

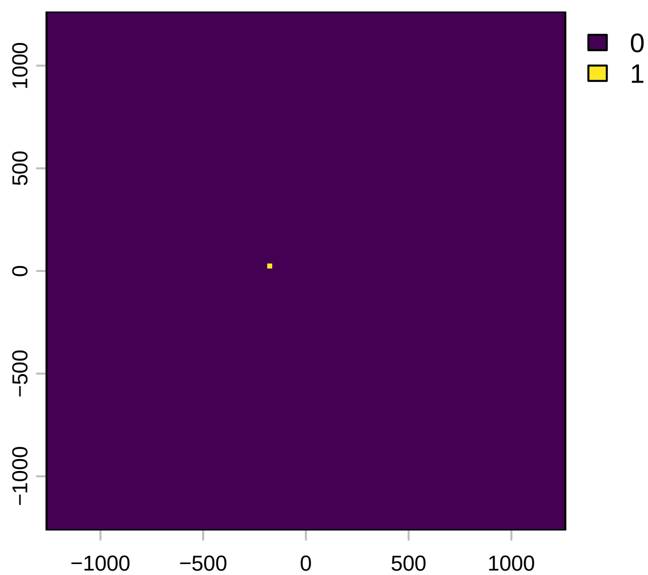
```
Warning: Unknown or uninitialised column: `pres_cent_x1`.
```

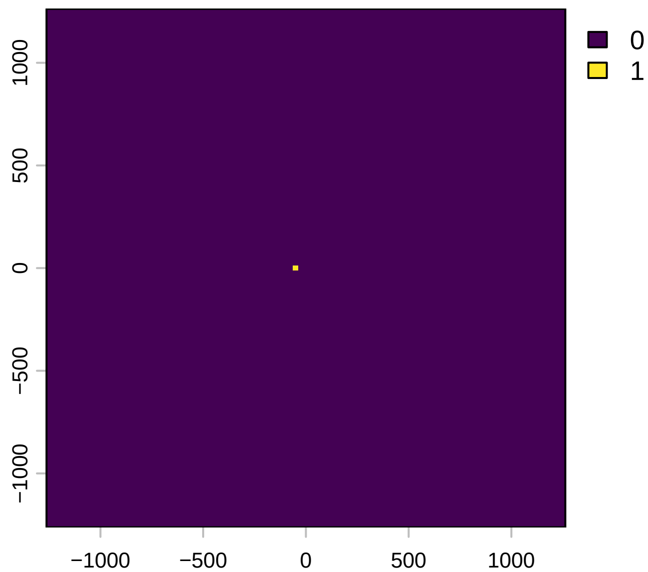
```
# Target (location of the next step)
walk(buffalo_data_covs$pres_cent_x2_within_cell[1:n_plots], terra::plot)
```


0
1
1000
500
0
−500
−1000
−1000
−500
0
500
1000


0
1
1000
500
0
−500
−1000
−1000
−500
0
500
1000


0
1
1000
500
0
−500
−1000
−1000
−500
0
500
1000


0
1
1000
500
0
−500
−1000
−1000
−500
0
500
1000



To save the object with all of the local covariates

```
# saveRDS(buffalo_data_covs, paste0("buffalo_data_id_covs_", Sys.Date(), ".rds"))
```

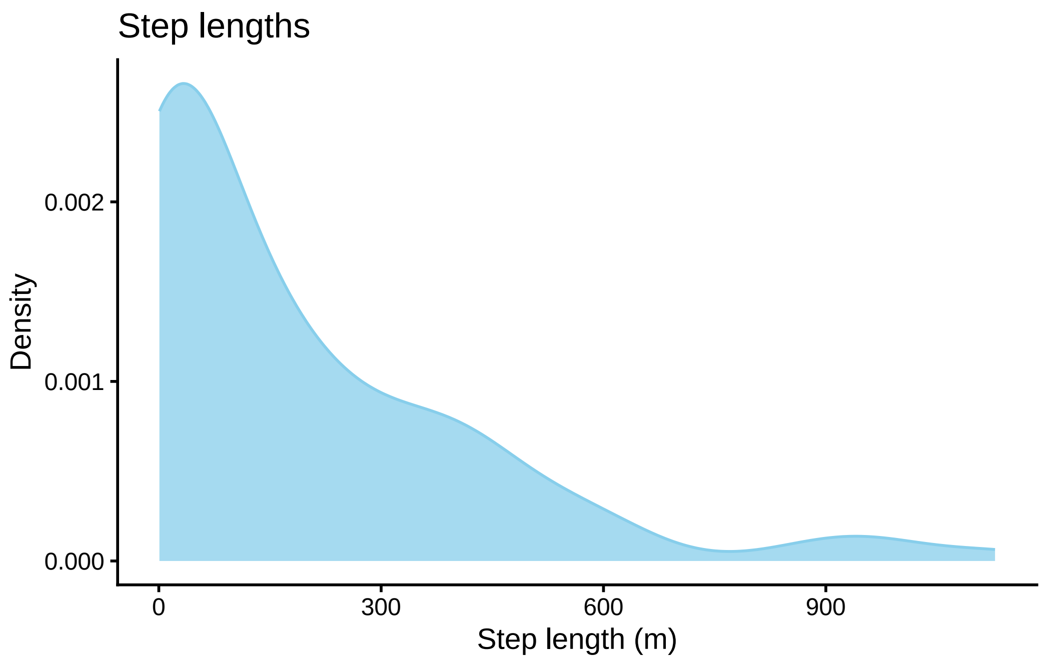
To save some plots

```
# for(i in 1:n_plots) {
#   png(filename = paste0("ndvi_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$ndvi_cent[[i]])
#   dev.off()
# }
#
# # for the pres layers
# for(i in 1:n_plots) {
#
#   # change the 0 values to NA
#   layer <- buffalo_data_covs$pres_cent[[i]]
#   layer[layer == 0] <- 0.5
#
#   ndvi_layer <- buffalo_data_covs$ndvi_cent[[i]]
#   new_layer <- layer*ndvi_layer
#
#   # save the plot
```

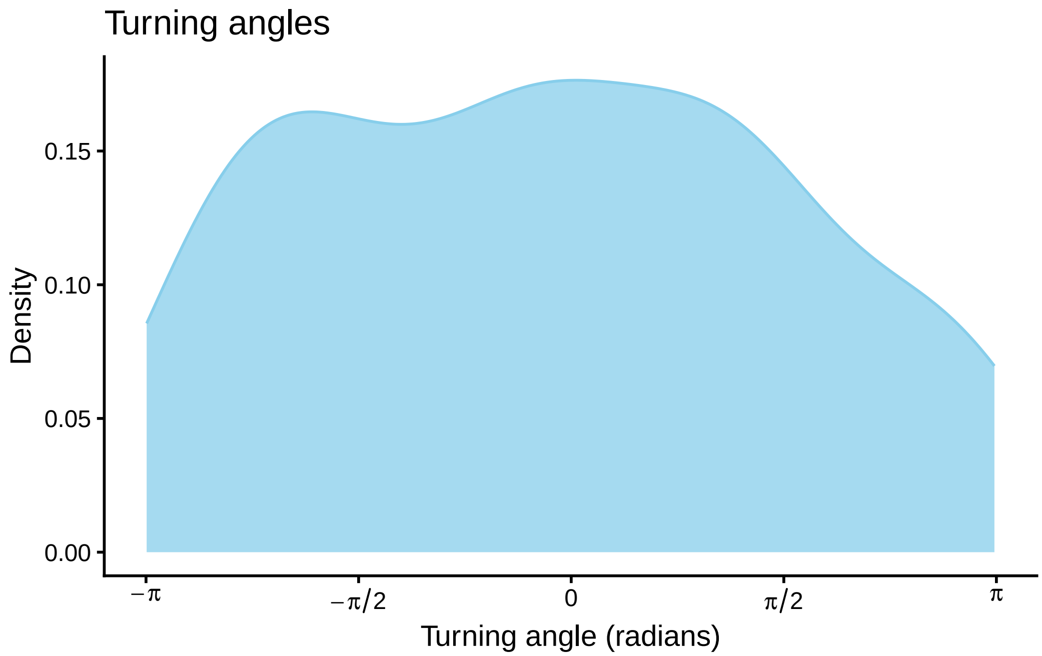
```
#   png(filename = paste0("pres_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(new_layer)
#   dev.off()
# }
#
#
# # other plots
#
# n_plots <- 1
#
# for(i in 1:n_plots) {
#   png(filename = paste0("veg_herby_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$veg_herby_cent[[i]])
#   dev.off()
# }
#
# for(i in 1:n_plots) {
#   png(filename = paste0("canopy_cover_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$canopy_cover_cent[[i]])
#   dev.off()
# }
#
# for(i in 1:n_plots) {
#   png(filename = paste0("slope_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$slope_cent[[i]])
#   dev.off()
# }
```

Plot the step lengths and turning angles

```
# plot step lengths
buffalo_data_covs %>% ggplot() +
  geom_density(aes(x = sl),
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  scale_x_continuous("Step length (m)") +
  scale_y_continuous("Density") +
  ggtitle("Step lengths") +
  theme_classic()
```




```
# ggsave("outputs/step_length.png", height = 60, width = 120, units = "mm", dpi = 600)


# plot turning angles
buffalo_data_covs %>% ggplot() +
  geom_density(aes(x = ta),
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  scale_x_continuous("Turning angle (radians)",
    breaks = c(-pi, -pi/2, 0, pi/2, pi),
    labels = c(expression(-pi), expression(-pi/2), "0", expression(pi/2), expression(pi))
  ) +
  scale_y_continuous("Density") +
  ggtitle("Turning angles") +
  theme_classic()
```




```
# ggsave("outputs/turning_angle.png", height = 60, width = 120, units = "mm", dpi = 600)
```